

D298

Quality and Impact of Adult E-Learning Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional utility built a 90-minute safety module for line technicians who average fifteen years on the job. The module uses the company's new brand colors and was finished two weeks early. Analytics show most technicians click through the first 60 minutes of general background in under five minutes. What redesign best fits how these adults learn?

- A. Lock the navigation so no one can advance until each screen's timer finishes.
- B. Refresh the visuals so they match the brand colors across all screens.
- C. Open with realistic field problems and let technicians test out of background they already know.
- D. Adults always prefer shorter courses, so cut the module to ten minutes total.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A designer for a staffing firm with offices in four states builds a narrated animation of a hiring workflow. Every screen shows a paragraph of text, and the narrator reads that exact paragraph aloud while the diagram animates. The voice actor was hired through an agency. Pilot learners score lower than on an earlier version. What change is most likely to improve learning?

- A. Keep the narration and diagram, and remove the full on-screen paragraphs.
- B. Move the paragraphs into a downloadable PDF and drop the diagram.
- C. Keep the paragraphs and diagram, and add a second narrator for variety.
- D. Replace the agency voice actor with an internal employee.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A module for new retail supervisors states the objective: "Given a scheduling conflict, the supervisor will build a compliant weekly staff schedule." The final quiz asks learners to match ten labor-law terms to their definitions. The module has won an internal design award and uses branching video. Which quality issue is most important?

- A. The quiz is too short; it should have twenty matching terms instead of ten.
- B. There is no issue, because award-winning modules meet quality standards.
- C. The assessment is misaligned; it should have learners build a schedule from a conflict scenario.
- D. The objective should be rewritten as "understand labor law" to match the quiz.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An adult e-learning course is reviewed with a framework that requires 85% of 100 possible points and every essential standard met. The review team spent three weeks on it and included an outside reviewer. The course earns 93 points, but one essential standard on accessible media is marked not met. What is the review result?

- A. It does not yet meet the framework; the essential standard must be fixed first.
- B. It fails permanently and must be rebuilt from scratch.
- C. It meets the framework, because 93 is above the 85-point cutoff.
- D. It meets the framework, because an outside reviewer took part.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A course meets every essential standard in a quality framework that requires 85 of 100 points. What is the lowest total score that lets it meet the framework?

- A. 15 points
- B. 100 points, because every standard must be met.
- C. Any score, because essentials are all that matter.
- D. 85 points

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A learning dashboard for a compliance course shows average time on page rose from 22 to 41 minutes after a redesign, and logins doubled. The redesign added a new color theme and a mobile layout. There is no assessment data yet. What can the designer fairly claim?

- A. The mobile layout caused the gain in learning.
- B. The redesign failed, because longer time means the content is harder to understand.
- C. Learning nearly doubled, because time on page nearly doubled.
- D. Learner activity and effort increased; whether learning improved is not yet known.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A five-module onboarding course launched in a busy hiring season. Enrollment data: 1,000 started Module 1, 900 started Module 2, 540 started Module 3, 500 started Module 4, and 460 started Module 5. The course sponsor prefers video over text. Where should the designer look first for a problem?

- A. The move from Module 1 to Module 2, because 100 learners left.
- B. Module 5, because it has the fewest learners.
- C. The move from Module 2 to Module 3, where only 60% continued.
- D. Convert all text modules to video, as the sponsor suggests.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In an e-learning funnel, 800 learners start Lesson 1 and the Lesson 1-to-2 continuation rate is 75%. If Lesson 2-to-3 continuation is 80%, how many learners start Lesson 3?

- A. 640
- B. 440
- C. 600
- D. 480

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Two cohorts took a revised data-privacy module. Cohort A, from the call center, went from a 40 pre-test average to 70. Cohort B, from the legal team, went from 80 to 95. The legal team completed the module on tablets. Using normalized gain, which cohort realized more of its possible improvement?

- A. They cannot be compared because Cohort B used tablets.
- B. Cohort B, with a normalized gain of 0.75.
- C. Cohort A, because it gained 30 points versus 15.
- D. Cohort A, with a normalized gain of 0.75.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A group's pre-test average was 60 and its normalized gain was 0.50. What was its post-test average?

- A.** 50
- B.** 80
- C.** 90
- D.** 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital's training team reports that 94% of nurses rated a new e-learning module on infusion pumps "very useful." The module was built in a popular authoring tool and took four months. Leadership asks whether nurses are now using the pumps correctly on the units. Which evidence answers leadership's question?

- A.** Structured observations or pump logs of correct use on the units weeks after training.
- B.** The 94% "very useful" rating.
- C.** A review of how long the authoring tool took to build the module.
- D.** End-of-module quiz scores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An evaluator has data showing a 30% drop in pump-related incident reports across the hospital after training. Which evaluation level does this evidence most directly address?

- A.** Learning, because the drop proves every nurse mastered the content.
- B.** Results
- C.** Reaction
- D.** Behavior

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional utility built a 90-minute safety module for line technicians who average fifteen years on the job. The module ~~uses the company's new brand colors and was finished two weeks early.~~ Analytics show most technicians click through the first 60 minutes of general background in under five minutes. What redesign best fits how these adults learn?

Target word: redesign that matches adult learning needs **Decides:** Experienced adults are skipping generic background content.

A	Inverted operation	Forcing seat time increases control instead of learner direction and does not add relevance.
B	Camouflage	Answers the branding detail; the problem is relevance, not appearance.
C	Correct	Problem-centered, self-directed design respects adults' experience and ties content to their work.
D	Absolute language	Length alone is not the principle; relevance and control are, and some content may still be needed.

2. Answer A Concept 2

A designer for a staffing firm ~~with offices in four states~~ builds a narrated animation of a hiring workflow. Every screen shows a paragraph of text, and the narrator reads that exact paragraph aloud while the diagram animates. ~~The voice actor was hired through an agency.~~ Pilot learners score lower than on an earlier version. What change is most likely to improve learning?

Target word: fix for the redundancy problem **Decides:** Identical on-screen text and narration alongside a diagram.

A	Correct	Removing redundant printed text cuts extraneous load; words go to the ear and pictures to the eye.
B	Dropped element	Removes the visual that carries the workflow; the diagram plus narration is the effective pair.
C	Inverted operation	Adds more audio instead of removing the redundant channel.
D	Camouflage	Who records the voice does not decide the redundancy issue.

3. Answer C Concept 3

A module for new retail supervisors states the objective: "Given a scheduling conflict, the supervisor will build a compliant weekly staff schedule." The final quiz asks learners to match ten labor-law terms to their definitions. ~~The module has won an internal design award and uses branching video.~~ Which quality issue is most important?

Target word: biggest quality problem **Decides:** The quiz checks term recall while the objective requires building a schedule.

A	Adjacent category	More recall items do not fix a mismatch in the kind of performance.
B	Camouflage	An award and video do not show alignment.
C	Correct	Assessment must measure the performance the objective names.
D	Inverted operation	Lowers the objective to a vague, unmeasurable verb instead of fixing the assessment.

4. Answer A Concept 4

An adult e-learning course is reviewed with a framework that requires 85% of 100 possible points and every essential standard met. ~~The review team spent three weeks on it and included an outside reviewer.~~ The course earns 93 points, but one essential standard on accessible media is marked not met. What is the review result?

Target word: pass or not under a threshold framework **Decides:** One essential standard is not met.

A	Correct	Essential standards are required regardless of the point total.
B	Absolute language	A missed standard is revised and re-reviewed, not a permanent failure.
C	Dropped element	Ignores the second condition: every essential standard must be met.
D	Camouflage	Who reviewed does not change the result.

5. Answer D Concept 4 • transfer

A course meets every essential standard in a quality framework that requires 85 of 100 points. What is the lowest total score that lets it meet the framework?

Target word: solve for the cutoff when essentials are met **Decides:** Both conditions: essentials met and at least 85 points.

A	Inverted operation	That is the number of points a course can miss, not the minimum score.
B	Confusion set	Confuses essential standards with all standards.
C	Dropped element	Drops the point threshold.
D	Correct	With essentials met, the point threshold is the only remaining condition.

6. Answer D Concept 5

A learning dashboard for a compliance course shows average time on page rose from 22 to 41 minutes after a redesign, and logins doubled. The redesign ~~added a new color theme and a mobile layout.~~ There is no assessment data yet. What can the designer fairly claim?

Target word: valid claim from the data given **Decides:** Only activity data exist; no assessment evidence.

A	Camouflage	Assumes a learning gain and a cause from design details.
B	Absolute language	Longer time is ambiguous; it cannot prove failure any more than success.
C	Confusion set	Treats effort data as learning data.
D	Correct	Time and logins measure engagement and effort, not learning.

7. Answer C Concept 6

A five-module onboarding course ~~launched in a busy hiring season.~~ Enrollment data: 1,000 started Module 1, 900 started Module 2, 540 started Module 3, 500 started Module 4, and 460 started Module 5. ~~The course sponsor prefers video over text.~~ Where should the designer look first for a problem?

Target word: largest step-to-step drop **Decides:** Retention from Module 2 to 3 is $540/900 = 60\%$, the lowest rate.

A	Wrong stem number	360 left between Modules 2 and 3, a far bigger loss.
B	Confusion set	The last step always has the smallest count; the rate matters.
C	Correct	Rates: 90%, 60%, about 93%, 92%; the 2→3 step loses the most.
D	Camouflage	Acts on a preference, not on the drop-off data.

8. Answer D Concept 6 · transfer

In an e-learning funnel, 800 learners start Lesson 1 and the Lesson 1-to-2 continuation rate is 75%. If Lesson 2-to-3 continuation is 80%, how many learners start Lesson 3?

Target word: solve for a count from rates **Decides:** Multiply the rates in sequence.

A	Wrong stem number	Applies only the 80% rate to the starting 800.
B	Inverted operation	Subtracts 45 percentage points from 800 in one step (800×0.55) instead of multiplying the rates.
C	Dropped element	Stops after the first step.
D	Correct	$800 \times 0.75 = 600$; $600 \times 0.80 = 480$.

9. Answer B Concept 7

Two cohorts took a revised data-privacy module. Cohort A, from the call center, went from a 40 pre-test average to 70. Cohort B, from the legal team, went from 80 to 95. ~~The legal team completed the module on tablets.~~ Using normalized gain, which cohort realized more of its possible improvement?

Target word: compare normalized gains **Decides:** A: $30/60 = 0.50$; B: $15/20 = 0.75$.

A	Camouflage	Device does not block the calculation.
B	Correct	$(95 - 80)/(100 - 80) = 0.75$ vs. $(70 - 40)/(100 - 40) = 0.50$.
C	Dropped element	Uses raw gain and drops the room-to-grow denominator.
D	Inverted operation	Divides A's gain by its pre-score ($30/40$) instead of by $100 - \text{pre}$.

10. Answer B Concept 7 · transfer

A group's pre-test average was 60 and its normalized gain was 0.50. What was its post-test average?

Target word: solve for post-test **Decides:** $\text{post} = \text{pre} + g \times (100 - \text{pre})$.

A	Wrong stem number	Reads the 0.50 gain as a score.
B	Correct	$60 + 0.50 \times (100 - 60) = 60 + 20 = 80$.
C	Inverted operation	Applies the gain to the pre-score (60×1.5) instead of to the room left to grow.
D	Dropped element	That is the raw point gain; the pre-score still has to be added.

11. Answer A Concept 8

A hospital's training team reports that 94% of nurses rated a new e-learning module on infusion pumps "very useful." The module was ~~built in a popular authoring tool and took four months.~~ Leadership asks whether nurses are now using the pumps correctly on the units. Which evidence answers leadership's question?

Target word: evidence for on-the-job behavior **Decides:** The question is about use on the job (behavior level).

A	Correct	Behavior-level evidence shows transfer to the job.
B	Adjacent category	That is reaction-level data.
C	Camouflage	Development effort says nothing about impact.
D	Confusion set	Shows learning, not whether nurses apply it on the units.

12. Answer B Concept 8 · transfer

An evaluator has data showing a 30% drop in pump-related incident reports across the hospital after training. Which evaluation level does this evidence most directly address?

Target word: classify the evidence by level **Decides:** An organizational outcome is results-level evidence.

A	Absolute language	Outcome data cannot prove individual mastery for every nurse.
B	Correct	Incident rates are organizational outcomes.
C	Confusion set	Reaction is satisfaction with the training.
D	Adjacent category	Behavior is individual on-the-job use; incident rates are the outcome of that use.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Adult learners need relevance	Adults engage when learning is problem-centered, tied to their real work, and lets them direct their own path.	Treating adults like a captive audience that will sit through content because it is assigned.	Experienced adults skipping or rushing generic content.
2	Redundancy and extraneous load	Narrating a graphic while showing the same words on screen adds extraneous load; narration plus the graphic usually teaches better.	Thinking more channels of the same words always help learning.	Full on-screen text read word for word by a narrator.
3	Alignment as the core quality test	A quality e-learning solution aligns objectives, activities, and assessments so each assessment measures what an objective states.	Judging quality by polish, media, or length.	An assessment that checks something the objective never asked for.
4	Essential standards in a quality rubric	In a threshold quality framework, a course must meet every essential standard and reach the point cutoff; a high total does not offset a missed essential.	Adding up points and ignoring which standards are required.	A strong total score with one essential standard marked not met.
5	Engagement data is not learning data	Clicks, logins, and time on page show activity or effort; only assessment evidence shows learning.	Reading high time-on-page as proof people learned.	A dashboard with activity counts and a claim about learning.
6	Finding the drop-off point	Compare step-to-step retention rates in a funnel; the step with the lowest rate is where to redesign first.	Picking the step with the smallest raw count, which is always the last step.	A sequence of counts for modules or screens.
7	Normalized learning gain	Normalized gain = (post – pre) / (100 – pre); it shows how much of the possible improvement was achieved.	Comparing raw point gains across groups that started at different levels.	Two groups with different pre-test averages.
8	Levels of impact evaluation	Reaction (satisfaction), learning (knowledge/skill), behavior (on-the-job use), results (organizational outcomes) are distinct levels needing different evidence.	Using satisfaction surveys as proof of behavior change or results.	A question about which evidence shows a specific level of impact.

Course scope (public description): Quality and Impact of Adult E-Learning Solutions is the second of two courses in the Adult Learning Designer pathway. This course introduces the issues learning experience designers often encounter when designing e-learning experiences for adults. It also directs learners to resources about current research related to issues and innovations in designing online learning experiences for adults. This course also outlines a quality framework for evaluating e-learning solutions for adults and provides opportunities for learners to apply that framework. Lastly, this course provides examples of how learning analytics can be used to determine the impact of e-learning solutions for adults. Learners will analyze dashboard data and determine the impact an e-learning solution has had on learner engagement, effort, and learning and how insights gained from data about learners and the learning experience can be used to optimize learning and the environments in which it occurs. Designing E-Learning Experiences for Adults is a prerequisite for this course.